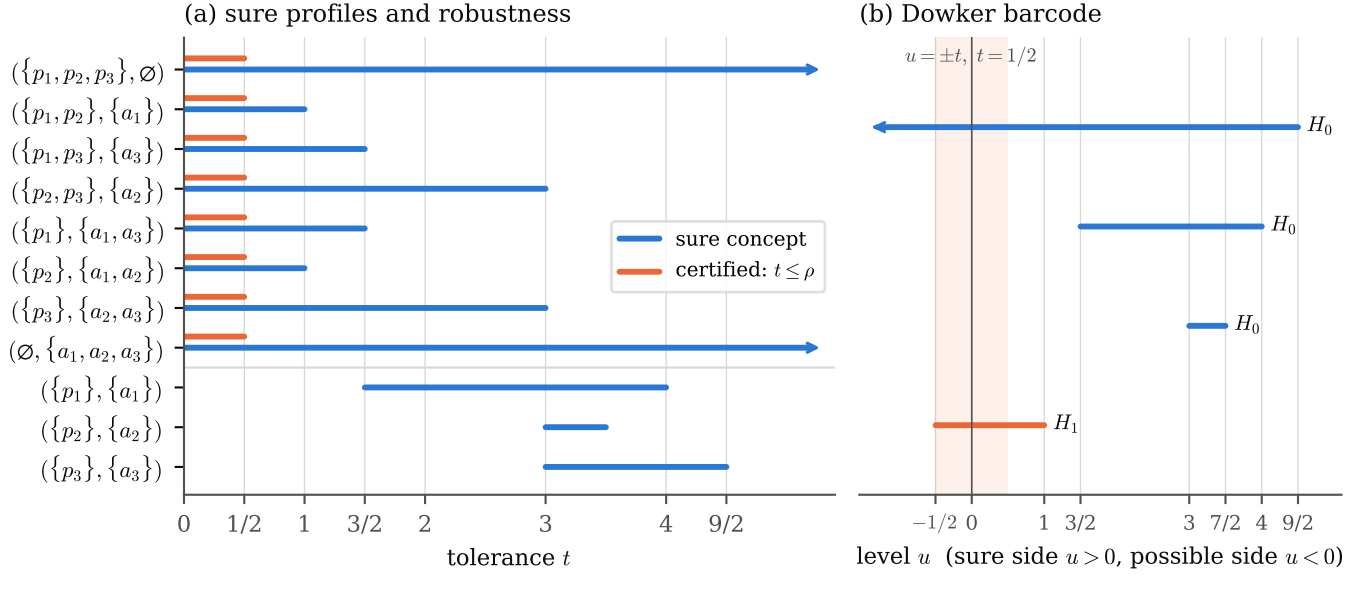
(a) sure profiles and robustness
$(\{p_1,p_2,p_3\},\varnothing)$
$(\{p_1,p_2\},\{a_1\})$
$(\{p_1,p_3\},\{a_3\})$
$(\{p_2,p_3\},\{a_2\})$
$(\{p_1\},\{a_1,a_3\})$
$(\{p_2\},\{a_1,a_2\})$
$(\{p_3\},\{a_2,a_3\})$
$(\varnothing,\{a_1,a_2,a_3\})$
$(\{p_1\},\{a_1\})$
$(\{p_2\},\{a_2\})$
$(\{p_3\},\{a_3\})$
sure concept
certified: $t \le \rho$
0 1/2 1 3/2 2 3 4 9/2
tolerance t
(b) Dowker barcode
$u = \pm t,\ t = 1/2$
H_0
H_0
H_0
H_1
−1/2 0 1 3/2 3 7/2 4 9/2
level u (sure side $u > 0$, possible side $u < 0$)